



香港中文大學(深圳)

The Chinese University of Hong Kong, Shenzhen

Introduction to Data Science

Lecture 23 Unsupervised learning: Clustering

Unsupervised Learning

Unsupervised Learning

- Data lacks structured or objective answers, such as labels.
- In other words, for all samples (x^i, y^i) , where $i = 1, \dots, N$, you can observe x^i but y^i remains unseen.

Training data



~~$y=1$ (cat)~~



~~$y=0$ (dog)~~



~~$y=1$ (cat)~~

...



~~$y=0$ (dog)~~

What Is Unsupervised Machine Learning?

- There is no predefined correct output for a given input.
- Instead, the algorithm must interpret the input and make the appropriate decision.
- The aim is to **examine the data and discern underlying patterns.**

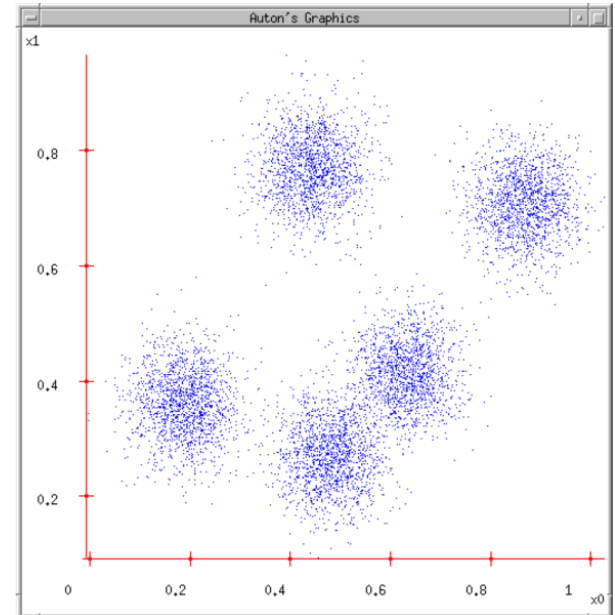
Examples

- The algorithm can identify customer segments who possess similar attributes. Customers within these segments can then be targeted by similar marketing campaigns.
- The algorithms are subsequently used to segment topics, identify outliers and recommend items.

Clustering

So, what is clustering in general?

- The algorithm figures out the grouping of objects based on the chosen **similarity/dissimilarity function**
 - **Points within a cluster are similar**
 - **Points across clusters are not so similar**



How to conduct?



Introduce a similarity function to represent whether two objects are similar.

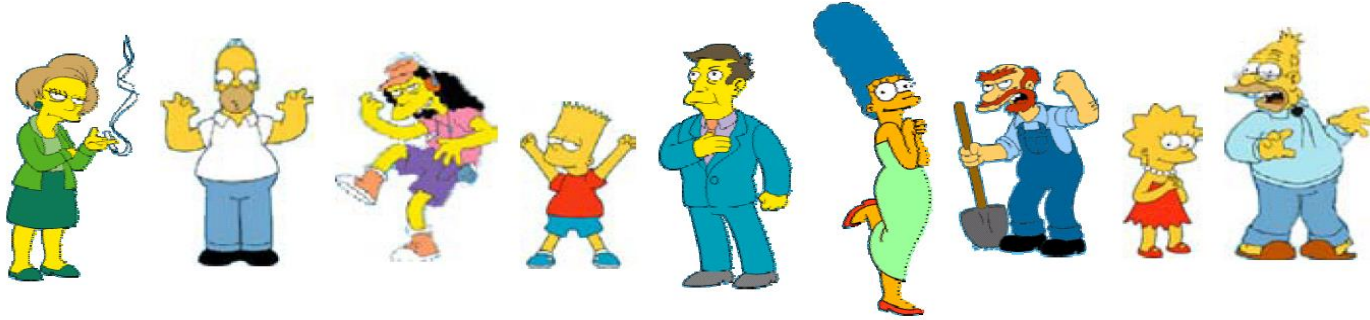


Divide objects into groups.

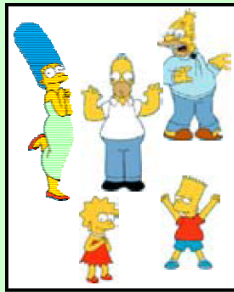
Do these two pictures exhibit similarity?



There is no universal standard for clustering



Clustering is subjective



Simpson's Family



School Employees



Females



Males

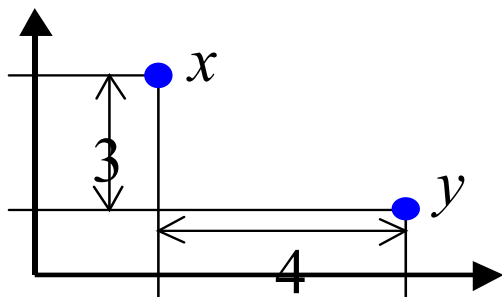
How to develop a dissimilarity/similarity function?

- Desired properties of dissimilarity functions
 - Symmetry: $d(x, y) = d(y, x)$
 - *Otherwise you could claim "Alex looks like Bob, but Bob looks nothing like Alex"*
 - Positive separability: $d(x, y) = 0$, if and only if $x = y$
 - *Otherwise there are objects that are different, but you cannot tell apart*
 - Triangular inequality: $d(x, y) \leq d(x, z) + d(z, y)$
 - *Otherwise you could claim "Alex is very like Bob, and Alex is very like Carl, but Bob is very unlike Carl"*

Distance functions for vectors

- Given two data points, both in R^n
 - $x = (x_1, x_2, \dots, x_n)^\top$
 - $y = (y_1, y_2, \dots, y_n)^\top$
- Euclidian distance: $d(x, y) = \sqrt{\sum_{i=1}^n (x_i - y_i)^2}$
- Minkowski distance: $d(x, y) = \sqrt[p]{\sum_{i=1}^n (x_i - y_i)^p}$
 - Euclidian distance: $p = 2$
 - Manhattan distance: $p = 1, d(x, y) = \sum_{i=1}^n |x_i - y_i|$
 - “inf”-distance: $p = \infty, d(x, y) = \max_{i=1}^n |x_i - y_i|$

Distance example



- Euclidian distance: $\sqrt{4^2 + 3^2} = 5$
- Manhattan distance: $4 + 3 = 7$
- “inf”-distance: $\max\{4, 3\} = 4$

How to conduct?



Introduce a similarity function to represent whether two objects are similar.



Divide objects into groups.

K-Means Clustering

K-means

- Given m data points, $\{x^1, x^2, \dots, x^m\}$
- Find k cluster centers, $\{c^1, c^2, \dots, c^k\}$
- And assign each data point i to one cluster, $\pi(i) \in \{1, \dots, k\}$
- Such that the sum of the distances from each data point to its respective cluster center is minimized

$$\min_{c, \pi} \sum_{i=1}^m d(x^i, c^{\pi(i)})$$

L2-norm distance

- Given m data points, $\{x^1, x^2, \dots, x^m\}$
- Find k cluster **centers**, $\{c^1, c^2, \dots, c^k\}$
- And assign each data point i to one cluster, $\pi(i) \in \{1, \dots, k\}$
- Such that the averaged distances from each data point to its respective cluster center is minimized

$$\min_{c, \pi} \frac{1}{m} \sum_{i=1}^m \|x^i - c^{\pi(i)}\|^2$$

K-means algorithm

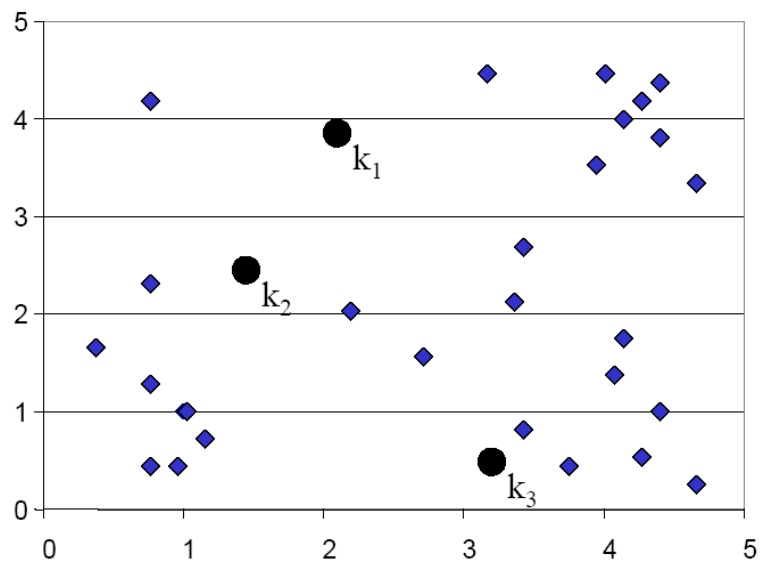
- Step 1: Initialize k cluster centers, $\{c^1, c^2, \dots, c^k\}$, randomly
- Step 2: Do
 - Decide the cluster memberships of each data point, x^i , by assigning it to the nearest cluster center (**cluster assignment**)

$$\pi(i) = \underset{j=1, \dots, k}{\operatorname{argmin}} \|x^i - c^j\|^2$$

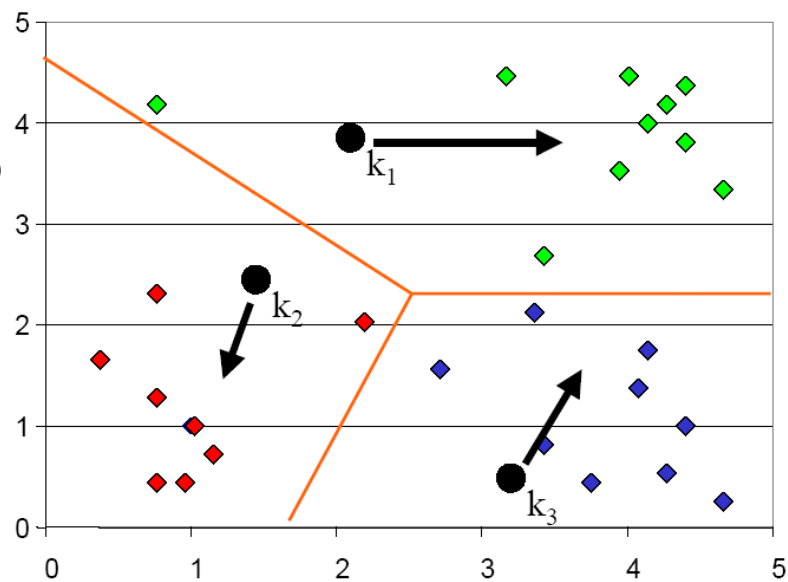
- Adjust the cluster centers (**center adjustment**)

$$c^j = \frac{1}{|\{i: \pi(i) = j\}|} \sum_{i: \pi(i) = j} x^i$$

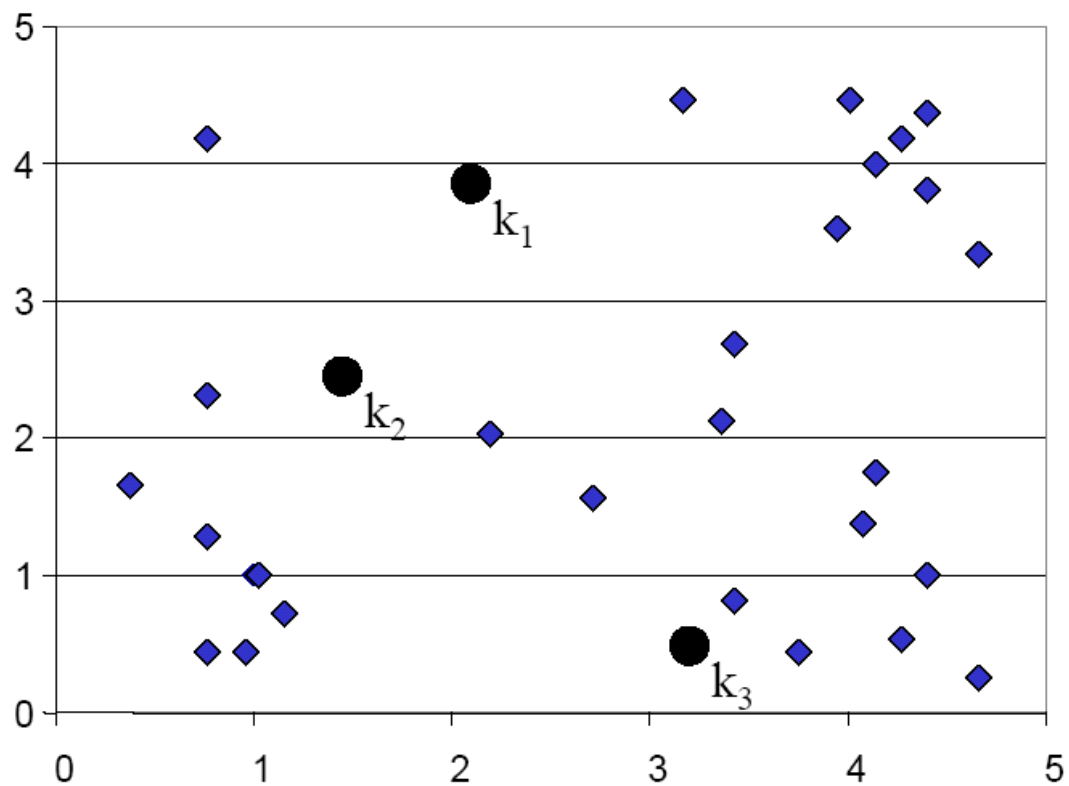
- While any cluster center undergoes changes, go to Step 2



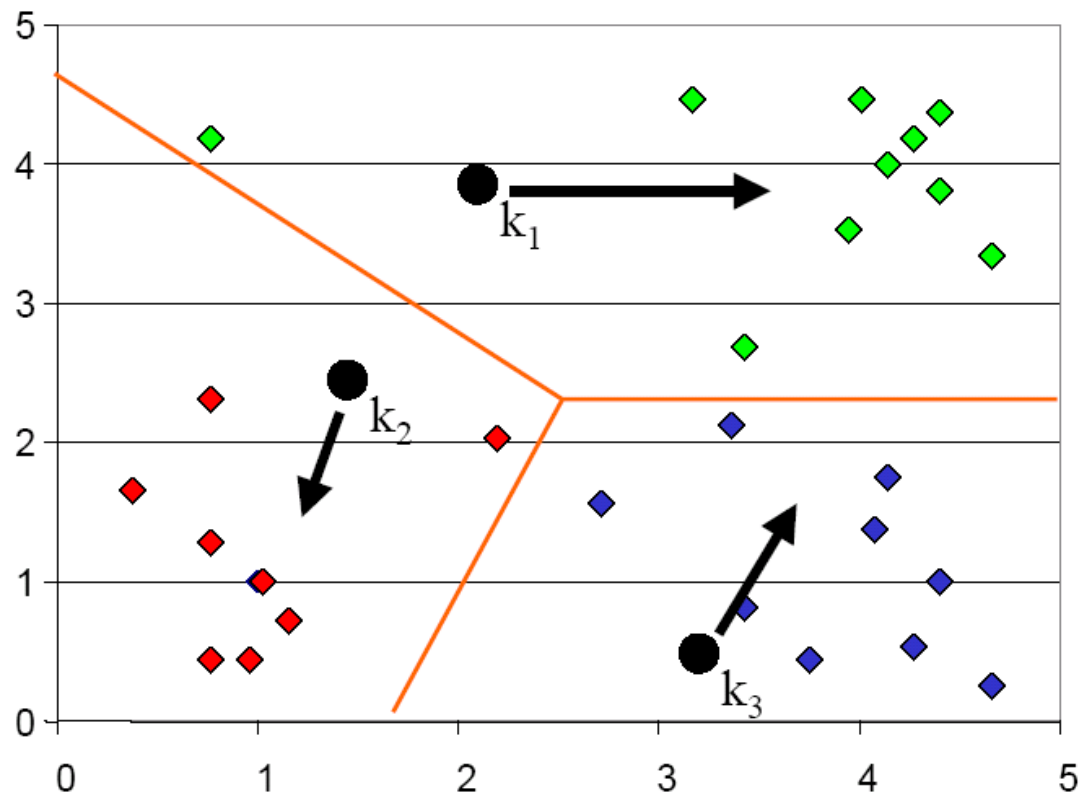
assign each
data point to
one cluster



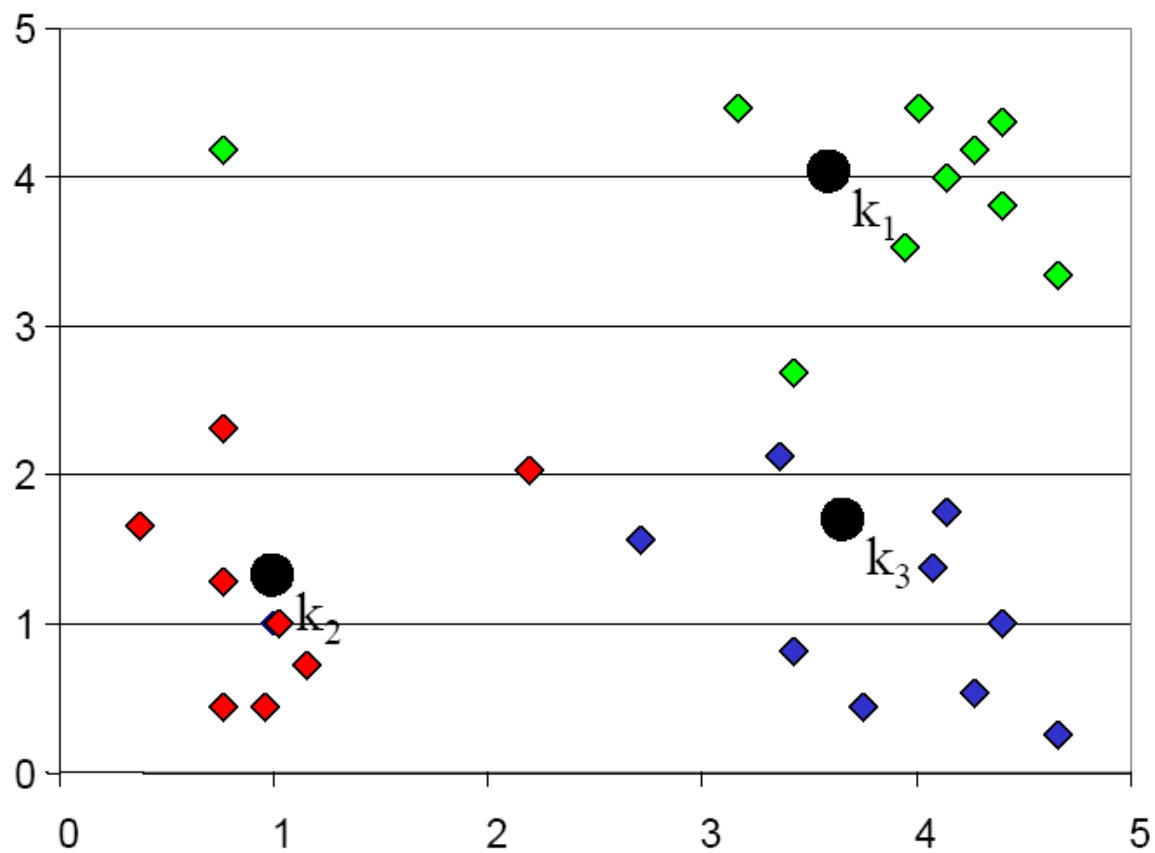
K-means: step 1



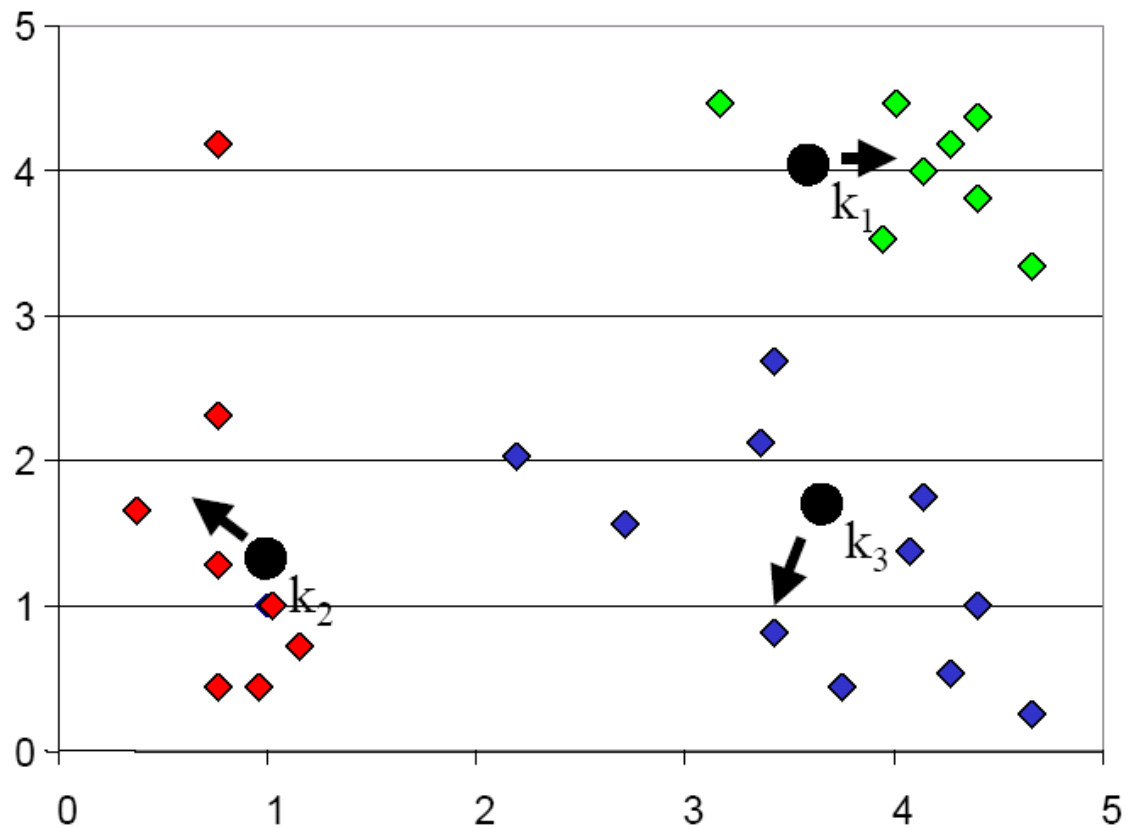
K-means: step 2



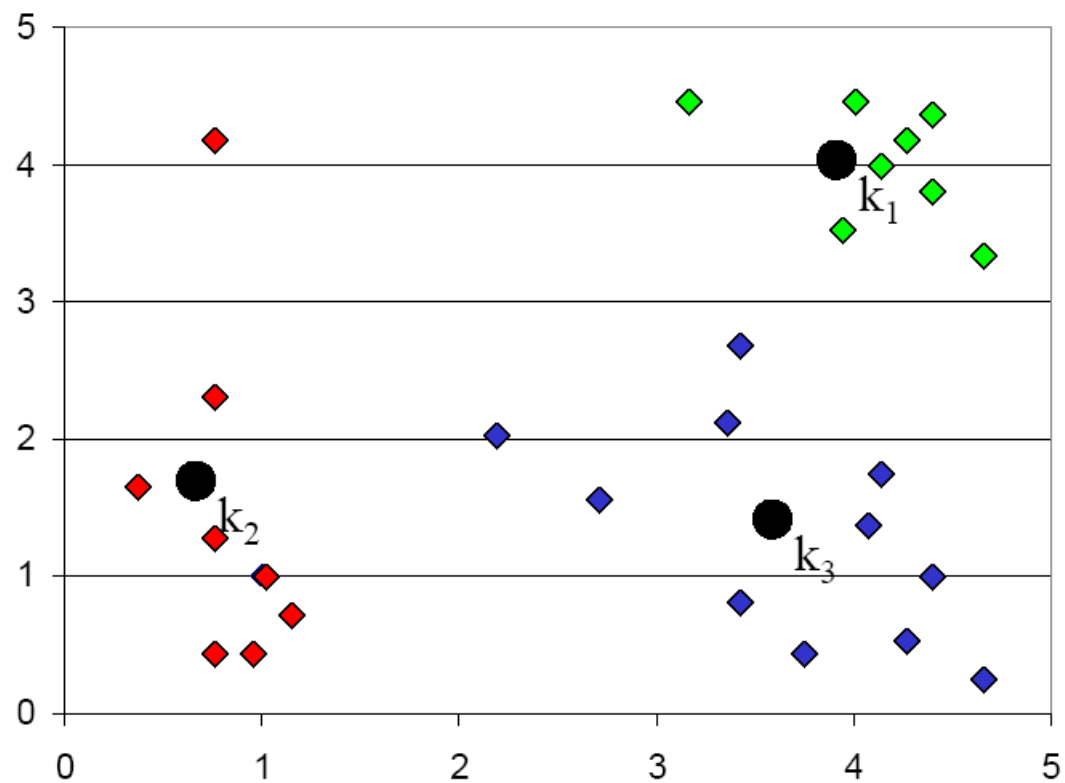
K-means: step 3



K-means: step 4



K-means: step 5



Questions

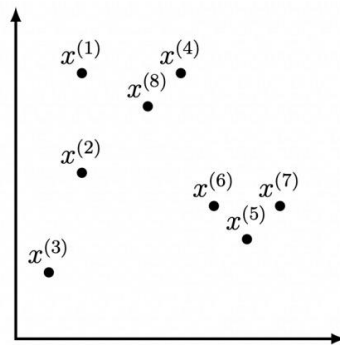
- Will different initialization lead to different results?
 - Yes
 - No

Exercise

[12 points] Suppose we would like to use the K-means algorithm and L2-norm distance (Euclidian distance) to cluster the 8 data points given in Figure 3 below into $K = 3$ clusters. The L2-norm distance between points $\mathbf{x} = (x_1, x_2)$ and $\mathbf{y} = (y_1, y_2)$ is $d(\mathbf{x}, \mathbf{y}) = \sqrt{(x_1 - y_1)^2 + (x_2 - y_2)^2}$. The coordinates of the data points are:

$$\begin{array}{llll} x^1 = (2, 8) & x^2 = (2, 5) & x^3 = (1, 2) & x^4 = (5, 8) \\ x^5 = (7, 3) & x^6 = (6, 4) & x^7 = (8, 4) & x^8 = (4, 7) \end{array}$$

Suppose x^1, x^3, x^5 are chosen as the initial cluster centers. Report the coordinates of the updated cluster centers in the next step and assign each data point according to the new cluster centers.



Exercise

$$d(x^1, c^1) = 0$$

$$d(x^2, c^1) = 3$$

$$d(x^3, c^1) = \sqrt{37}$$

$$d(x^4, c^1) = 3$$

$$d(x^5, c^1) = \sqrt{50}$$

$$d(x^6, c^1) = 32$$

$$d(x^7, c^1) = \sqrt{52}$$

$$d(x^8, c^1) = \sqrt{5}$$

$$d(x^1, c^2) = \sqrt{37}$$

$$d(x^2, c^2) = \sqrt{10}$$

$$d(x^3, c^2) = 0$$

$$d(x^4, c^2) = \sqrt{52}$$

$$d(x^5, c^2) = \sqrt{37}$$

$$d(x^6, c^2) = \sqrt{29}$$

$$d(x^7, c^2) = \sqrt{53}$$

$$d(x^8, c^2) = \sqrt{34}$$

$$d(x^1, c^3) = \sqrt{50}$$

$$d(x^2, c^3) = \sqrt{29}$$

$$d(x^3, c^3) = \sqrt{37}$$

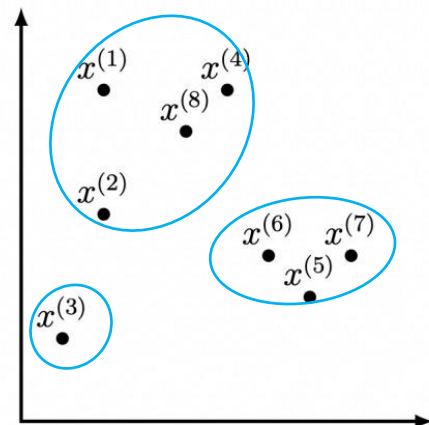
$$d(x^4, c^3) = \sqrt{29}$$

$$d(x^5, c^3) = 0$$

$$d(x^6, c^3) = \sqrt{2}$$

$$d(x^7, c^3) = \sqrt{2}$$

$$d(x^8, c^3) = 5$$



New centers: $(13/7, 7)$

$(1, 2)$

$(7, 11/3)$